

# 4.1 Analytical Techniques

## Question Paper

|            |                           |
|------------|---------------------------|
| Course     | CIEA Level Chemistry      |
| Section    | 4. Analysis               |
| Topic      | 4.1 Analytical Techniques |
| Difficulty | Hard                      |

**Time allowed:** 60

**Score:** /46

**Percentage:** /100

### Question 1a

This question is about the analysis of alcohols.

Alcohol **A** can be prepared by hydrolysing the halogenoalkane  $\text{C}_2\text{H}_5\text{CHBrCH}_3$  with aqueous sodium hydroxide. The melting and boiling point information of alcohol **A** and  $\text{C}_2\text{H}_5\text{CHBrCH}_3$  are given in Table 1.1.

Table 1.1

|                                       | melting point / K | boiling point / K |
|---------------------------------------|-------------------|-------------------|
| alcohol <b>A</b>                      | 158               | 373               |
| $\text{C}_2\text{H}_5\text{CHBrCH}_3$ | 161               | 364               |

Write an equation for the hydrolysis of  $\text{C}_2\text{H}_5\text{CHBrCH}_3$  with aqueous sodium hydroxide. Include state symbols in your answer

[2 marks]

### Question 1b

Using the information in Table 1.1, explain how alcohol **A** can be separated from the products identified in part (a).

[3 marks]

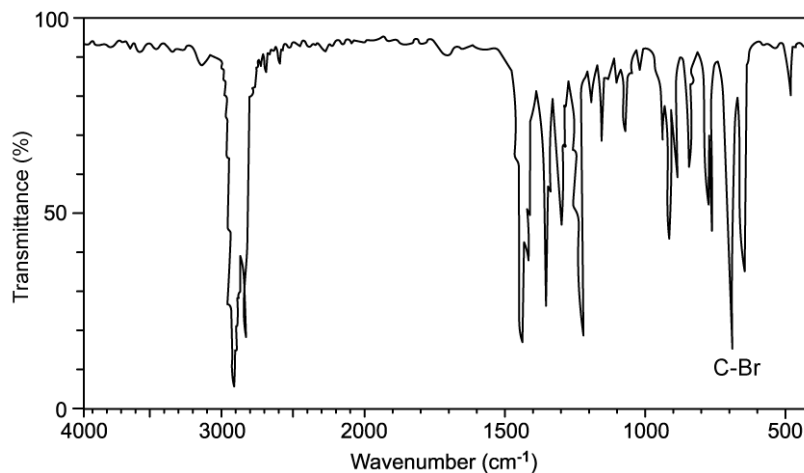
### Question 1c

Describe **two** chemical tests and their expected observations to confirm the identity of the inorganic product.

[2 marks]

## Question 1d

The infra-red spectrum of  $\text{C}_2\text{H}_5\text{CHBrCH}_3$  is shown in Fig. 1.1 with the C–Br bond absorption labelled.



**Fig. 1.1**

Using Fig. 1.1 and Table 1.2, predict **three** main differences that would be seen between the spectra of  $\text{C}_2\text{H}_5\text{CHBrCH}_3$  and alcohol **A**.

**Table 1.2**

| Bond                     | Functional groups containing the bond | Characteristic infrared absorption range (in wavenumber) / $\text{cm}^{-1}$ |
|--------------------------|---------------------------------------|-----------------------------------------------------------------------------|
| C–O                      | hydroxy, ester                        | 1040 – 1300                                                                 |
| C=C                      | aromatic compound, alkene             | 1500 – 1680                                                                 |
| C=O                      | amide                                 | 1640 – 1690                                                                 |
|                          | carbonyl, carboxyl                    | 1670 – 1740                                                                 |
|                          | ester                                 | 1710 – 1750                                                                 |
| $\text{C}\equiv\text{N}$ | nitrile                               | 2200 – 2250                                                                 |
| C–H                      | alkane                                | 2850 – 2950                                                                 |
| N–H                      | amine, amide                          | 3300 – 3500                                                                 |
| O–H                      | carboxyl                              | 2500 – 3000                                                                 |
|                          | hydroxy                               | 3200 – 3600                                                                 |

**[3 marks]**

### Question 1e

The mass spectrum of  $(\text{CH}_3)_2\text{CHCH}_2\text{OH}$  is shown in Fig. 1.2.

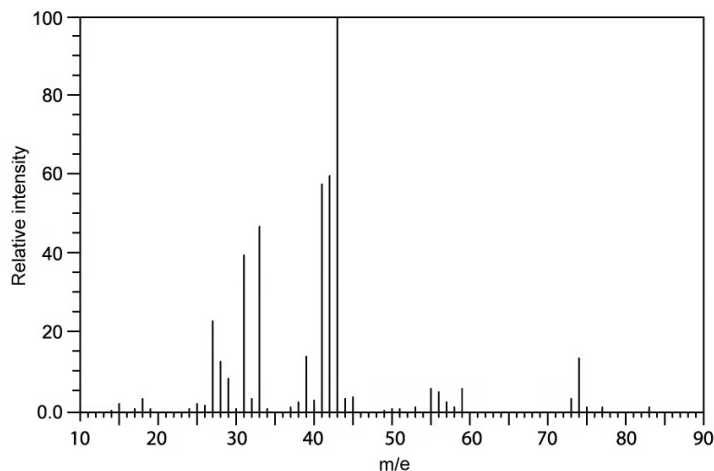


Fig. 1.2

Identify the ion that is responsible for the peak with the greatest relative intensity.

[1 mark]

### Question 2a

An unknown compound, **B**, is analysed and shown to have a composition of 69.7% carbon, 18.6% oxygen and the remainder is hydrogen.

The following **incomplete** mass spectrometry and IR spectroscopy data are collected.

#### Mass spectrum

- M+1 peak at  $m/e = 87.0$
- Fragment ion peak at  $m/e = 28$

#### Infra-red spectrum

- Sharp peak at  $1705\text{ cm}^{-1}$
- No broad peak around  $3200\text{ cm}^{-1}$

Calculate the molecular formula of compound **B**.

[3 marks]

**Question 2b**

Using Table 2.1 and the **incomplete** mass spectrometry and infra-red spectroscopy data, identify **two** possible isomers for compound **B**.

Table 2.1

| Bond | Functional groups containing the bond | Characteristic infrared absorption range (in wavenumber) / $\text{cm}^{-1}$ |
|------|---------------------------------------|-----------------------------------------------------------------------------|
| C–O  | hydroxy, ester                        | 1040 – 1300                                                                 |
| C=C  | aromatic compound, alkene             | 1500 – 1680                                                                 |
| C=O  | amide                                 | 1640 – 1690                                                                 |
|      | carbonyl, carboxyl                    | 1670 – 1740                                                                 |
|      | ester                                 | 1710 – 1750                                                                 |
| C≡N  | nitrile                               | 2200 – 2250                                                                 |
| C–H  | alkane                                | 2850 – 2950                                                                 |
| N–H  | amine, amide                          | 3300 – 3500                                                                 |
| O–H  | carboxyl                              | 2500 – 3000                                                                 |
|      | hydroxy                               | 3200 – 3600                                                                 |

Your answer should include an explanation proving your **two** possible isomers and disproving **one** other isomer.

[3 marks]

**Question 2c**

A mass spectrum peak at  $m/e = 29$  could be used to distinguish between the two remaining isomers.

Identify the fragment responsible for this peak and explain how it can be used to distinguish between the two isomers.

**[3 marks]**

### Question 3a

Unknown organic compound **C** is analysed by mass spectrometry and infra-red spectroscopy. The mass spectrum is shown in Fig. 3.1 and the infra-red spectrum is shown in Fig. 3.2.

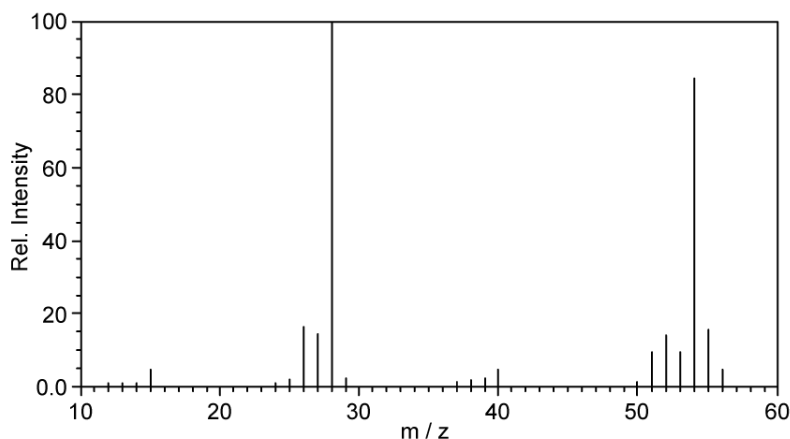


Fig 3.1

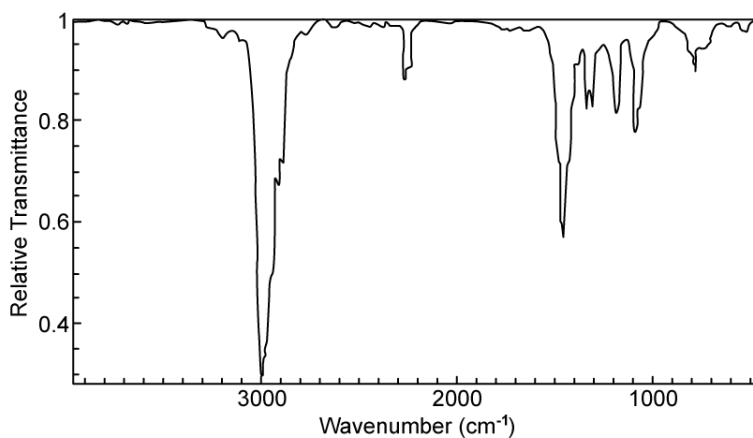


Fig 3.2

Deduce **one** possible identity for compound **C**. Explain your answer.

In your answer, use the molecular ion peak at  $m/e = 55$ , relevant infra-red absorptions in the region **above**  $1500\text{ cm}^{-1}$  and Table 3.1.

Table 3.1

| Bond | Functional groups containing the bond | Characteristic infrared absorption range (in wavenumber) / $\text{cm}^{-1}$ |
|------|---------------------------------------|-----------------------------------------------------------------------------|
| C–O  | hydroxy, ester                        | 1040 – 1300                                                                 |
| C=C  | aromatic compound, alkene             | 1500 – 1680                                                                 |

|     |                                      |                                           |
|-----|--------------------------------------|-------------------------------------------|
| C=O | amide<br>carbonyl, carboxyl<br>ester | 1640 – 1690<br>1670 – 1740<br>1710 – 1750 |
| C≡N | nitrile                              | 2200 – 2250                               |
| C–H | alkane                               | 2850 – 2950                               |
| N–H | amine, amide                         | 3300 – 3500                               |
| O–H | carboxyl<br>hydroxy                  | 2500 – 3000<br>3200 – 3600                |

[3 marks]

### Question 3b

The mass spectrum of compound **C** shows an  $M+1$  peak at  $m/e = 56$ .

Suggest **two** ways that this peak could be formed.

[1 mark]

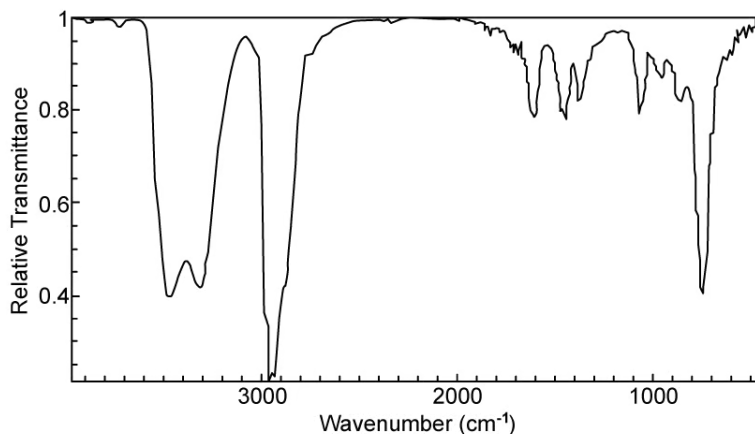
### Question 3c

Calculate the expected elemental percentage composition by mass of an 8.0 g sample of compound **C**. Give your answers to significant figures.

[2 marks]

### Question 3d

Compound **C** undergoes a reaction to form compound **D**. The IR spectrum of compound **D** is shown in Fig. 3.3.



**Fig. 3.3**

State the reagents required for this reaction. In your answer, use Table 3.1 and identify any relevant absorptions in the infra-red spectrum to justify your answer.

**[2 marks]**

### Question 4a

Alcohol **E** is analysed using a combination of techniques. The analytical data is shown below.

#### Elemental analysis by mass

- C = 68.2%, H = 13.6%, O = 18.2%

#### Mass spectrum

- Molecular ion peak at  $m/e = 88.0$
- Major ion peak at  $m/e = 45.0$

Determine the molecular formula of alcohol **E**.

**[3 marks]**

**Question 4b**

Alcohol **E** oxidises to form a carboxylic acid. Suggest what information this gives about the structure of alcohol **E**.

**[1 mark]****Question 4c**

Using the information from part **(b)**, draw the fully displayed formulae and give the systematic names of the **four** possible structural isomers that could be alcohol **E**.

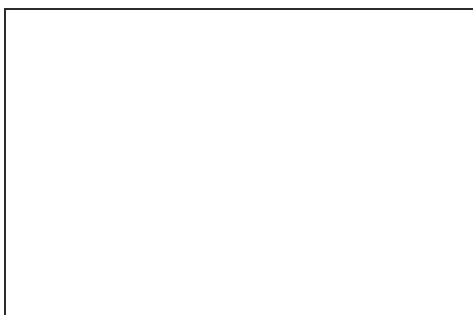
|            |            |
|------------|------------|
| Name ..... | Name ..... |
| Name ..... | Name ..... |

**[4 marks]**

**Question 4d**

The mass spectrum of alcohol **E** has a major peak at  $m/e = 45$ .

Draw the structure of the species that could give this peak.



[1 mark]

**Question 4e**

Alcohol **E** has a branched chain.

Identify the isomer from part **(d)** that is alcohol **E**. Explain your answer.

[1 mark]

### Question 5a

The term carbohydrate is often associated with sugars or suggests a compound that contains carbon, hydrogen and oxygen. It was previously used in chemistry to define any compound of the general formula  $C_m(H_2O)_n$ .

Compound **F** contains 6.98 % hydrogen and 37.21 % oxygen by mass. The remainder of the mass is carbon.

i)

Determine whether compound **F** is a carbohydrate according to the previous chemistry definition. Justify your answer.

[4]

ii)

Explain why this empirical formula cannot be the molecular formula of compound **F**. You should justify your answer in terms of hydrogen atoms and include a minimum of **three** displayed formulae.

[4]

[1 mark]

### Question 5b

Fig. 5.1 shows the mass spectrum of compound **F**.

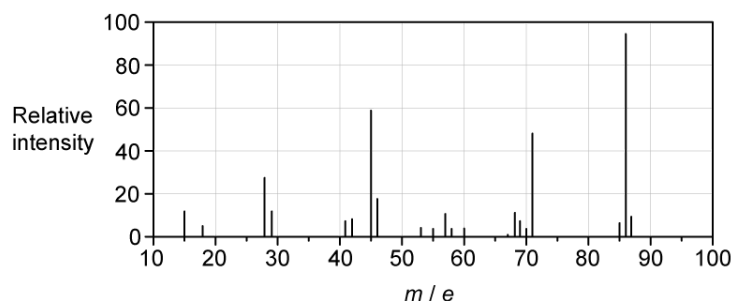


Fig. 5.1

Using your answer to part (a)(i), deduce the molecular formula of compound **F**.

[2 marks]



## Question 5c

Fig. 5.2 shows the infra-red spectrum of compound F.

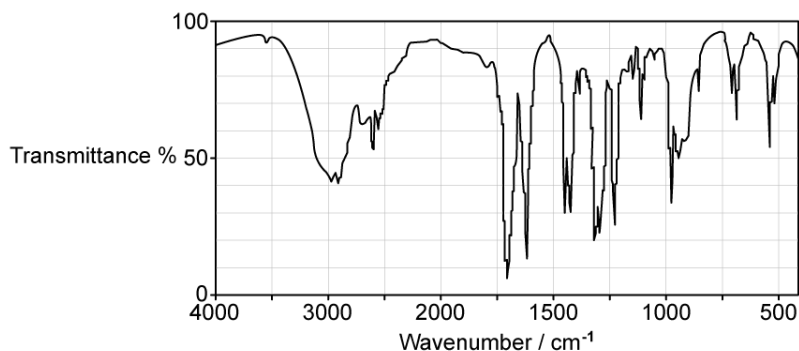


Fig. 5.2

Table 5.1 shows the characteristic infrared absorption frequencies for some selected bonds.

Table 5.1

| Bond | Functional groups containing the bond | Characteristic infrared absorption range (in wavenumber) / $\text{cm}^{-1}$ |
|------|---------------------------------------|-----------------------------------------------------------------------------|
| C–O  | hydroxy, ester                        | 1040 – 1300                                                                 |
| C=C  | aromatic compound, alkene             | 1500 – 1680                                                                 |
| C=O  | amide                                 | 1640 – 1690                                                                 |
|      | carbonyl, carboxyl                    | 1670 – 1740                                                                 |
|      | ester                                 | 1710 – 1750                                                                 |
| C≡N  | nitrile                               | 2200 – 2250                                                                 |
| C–H  | alkane                                | 2850 – 2950                                                                 |
| N–H  | amine, amide                          | 3300 – 3500                                                                 |
| O–H  | carboxyl                              | 2500 – 3000                                                                 |
|      | hydroxy                               | 3200 – 3600                                                                 |

Complete Table 5.2 with the bond responsible for each of the principal absorptions seen in the infra-red spectrum of compound F.

Table 5.2

| principal absorptions in infra-red spectrum | bond responsible |
|---------------------------------------------|------------------|
| 2500 - 3000 $\text{cm}^{-1}$                |                  |
| 1710 $\text{cm}^{-1}$                       |                  |

|                       |  |
|-----------------------|--|
| 1620 cm <sup>-1</sup> |  |
|-----------------------|--|

[1 mark]

**Question 5d**

Compound **F** is a straight-chain molecule.

Using all of the available information including your answers from parts **(b)** and **(c)**, draw the displayed formulae and give the systematic names of the **three** possible isomers that could be compound **F**.

|                       |                       |                       |
|-----------------------|-----------------------|-----------------------|
| <div>Name .....</div> | <div>Name .....</div> | <div>Name .....</div> |
|-----------------------|-----------------------|-----------------------|

[3 marks]

**Question 5e**

Compound **F** is the stereoisomer with the lowest boiling point.

Using the isomers identified in part **(d)**, give the name of compound **F**. Explain your answer.

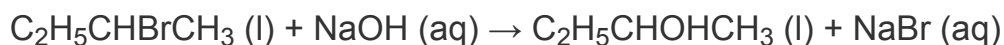
[1 mark]



## Model Answers: Hard

1a

a) The equation for the hydrolysis of  $\text{C}_2\text{H}_5\text{CHBrCH}_3$  with aqueous sodium hydroxide is:



- Correct chemical formulae; [1 mark]
- Correct state symbols; [1 mark]

[Total: 2 marks]

- The reaction that is taking place is the nucleophilic substitution of  $\text{C}_2\text{H}_5\text{CHBrCH}_3$  by  $\text{OH}^-$ 
  - $\text{C}_2\text{H}_5\text{CHBrCH}_3 + \text{NaOH} \rightarrow \text{C}_2\text{H}_5\text{CHOHCH}_3 + \text{NaBr}$
- You are given the state of NaOH in the question
- Using the boiling point information in the table, you should be able to deduce that alcohol **A** and  $\text{C}_2\text{H}_5\text{CHBrCH}_3$  are both liquids
- Sodium bromide is a soluble salt
  - $\text{C}_2\text{H}_5\text{CHBrCH}_3 (\text{l}) + \text{NaOH} (\text{aq}) \rightarrow \text{C}_2\text{H}_5\text{CHOHCH}_3 (\text{l}) + \text{NaBr} (\text{aq})$

1b

b) To separate alcohol **A** from the reaction products:

**EITHER (freezing method)**

- Distil the reaction mixture  
**AND**  
To separate the NaBr; [1 mark]
- Cool the mixture / distillate to (approximately)  $0^\circ\text{C}$   
**AND**  
To freeze / solidify the water (from the solution); [1 mark]
- Filter the mixture / distillate  
**AND**

To remove the solid water / ice; [1 mark]

**OR (separating funnel method)**

- Add an organic solvent (such as ether)

**AND**

To dissolve alcohol **A** / butan-2-ol; [1 mark]

- Use a separating funnel

**AND**

To separate the organic layer containing alcohol **A**; [1 mark]

- Distil the organic layer

**AND**

To remove the organic solvent / to separate alcohol **A** out of the organic solvent; [1 mark]

**[Total: 3 marks]**

- This question tests your knowledge of separation techniques
  - Some of this knowledge comes from your previous studies and some comes from experiments that you may have completed during this course
- There are 2 methods to separate alcohol **A** from the mixture of products:
  - Using freezing
  - Using a separating funnel
- The freezing method
  - The sodium bromide / NaBr is relatively easy to separate out
  - This is achieved by distilling the reaction mixture
    - But, the boiling point of alcohol **A** and water are the same which means that they will both distil off at the same time
  - You could, arguably, add a drying agent but there is likely to be a large volume of water and there is no guarantee that alcohol **A** will not react with the drying agent
  - Using the information in Table 1.1, the melting point of alcohol **A** is 115 ° lower than that of water
    - Therefore, cooling to 0 °C will cause the water to freeze
    - This means that the solid water is in liquid alcohol **A** and can be removed by filtration
- The separating funnel method
  - Uses an organic solvent to form an aqueous and an organic layer
    - The organic solvent should allow alcohol **A** to dissolve into it

- You can then separate the organic layer from the aqueous layer in a separating funnel
- This leaves you with alcohol **A** dissolved in an organic solvent
- The organic solvent can then be distilled off, leaving alcohol **A**

1c

c) **Two** chemical tests and their expected observations to confirm the identity of the inorganic product are:

Sodium ion /  $\text{Na}^+$ :

- Flame test / description of a flame test

**AND**

Yellow / orange flame; [1 mark]

Bromide ion /  $\text{Br}^-$ :

- Add acidified silver nitrate solution /  $\text{AgNO}_3(\text{aq})$  **OR** add nitric acid followed by silver nitrate solution /  $\text{AgNO}_3(\text{aq})$

**AND**

A cream precipitate forms; [1 mark]

**[Total: 2 marks]**

- **Careful:** The question asks you to confirm the identity of the inorganic product, which is sodium bromide
- Sodium bromide contains the sodium ion /  $\text{Na}^+$  and the bromide ion /  $\text{Br}^-$ 
  - Flame testing is used to identify the sodium ion /  $\text{Na}^+$ 
    - Although a question on flame testing is unlikely, it is still possible as this comes from prior knowledge
    - You need to either state "flame test" or describe how to perform a flame test
    - **Remember:** Sodium gives a yellow flame, although orange is usually an acceptable colour
  - The silver nitrate test is used to identify the bromide ion /  $\text{Br}^-$ 
    - The solution being tested needs to be acidified with nitric acid before silver nitrate solution /  $\text{AgNO}_3(\text{aq})$  is added
    - The bromide ion forms a cream precipitate of silver bromide

- You can carry on to explain how this precipitate is soluble in concentrated ammonia solution but not soluble in dilute ammonia solution, but this is not necessary for the mark

1d

d) **Three** main differences that would be seen between the spectra of  $\text{C}_2\text{H}_5\text{CHBrCH}_3$  and alcohol **A** are:

- The disappearance of the peak at  $500 - 800 \text{ cm}^{-1}$  / C–Br peak ; [1 mark]
- The appearance of a peak at  $3200 - 3600 \text{ cm}^{-1}$  / alcohol O–H peak; [1 mark]
- The appearance of a peak at  $1040 - 1300 \text{ cm}^{-1}$  / alcohol C–O peak; [1 mark]

[Total: 3 marks]

- With this question you need to read the question stem carefully
- The reaction that takes place is the nucleophilic substitution of a halogenoalkane by  $\text{OH}^-$  (sodium hydroxide) to form alcohol **A**, which means that:
  - The C–Br bond will be broken
  - Bromine will be substituted
  - A new bond will form between the  $\delta^+$  carbon and the  $\text{OH}^-$
  - There will also be an O–H present
- You need to comment on how the spectrum will change, including
  - The halogenoalkane peaks that will disappear
  - The alcohol peaks that will appear after the nucleophilic substitution reaction has occurred

1e

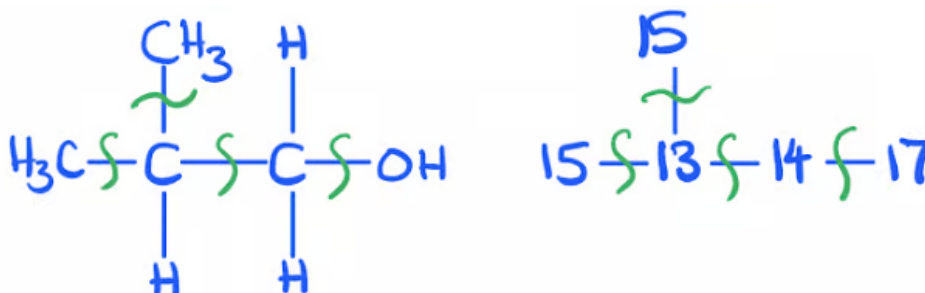
e) The ion that is responsible for the peak with the greatest relative intensity is:

- $(\text{CH}_3)_2\text{CH}^+$ ; [1 mark]

[Total: 1 mark]

- The peak with the greatest relative intensity has an  $m / e$  value of 43
- Try to draw the structures to see which were likely to be formed from  $(\text{CH}_3)_2\text{CHCH}_2\text{OH}$
- Other fragment ions that students have given as incorrect answers include:
  - $\text{CHCH}_2\text{OH}^+$ :  $m / e = 44$

- $\text{CH}_3\text{CH}_2\text{CH}^+$ :  $m/e = 42$
- Take your time to look at the overall formula of the molecule given,  $(\text{CH}_3)_2\text{CHCH}_2\text{OH}$ , and determine where the split could occur to form an ion with an  $m/e$  of 43
- **Remember:** you will lose a mark if you don't include the charge on your molecular ion
- Another method to identify ion peaks for molecules is:
  - Draw the whole molecule out
  - Break the major bonds between atoms as these will form the fragments



- You can even consider changing the groups for their mass value to help find the bigger fragments

2a

a) To calculate the molecular formula of compound **B**:

- Moles of carbon = 5.80

**AND**

Moles of oxygen = 1.16

**AND**

Moles of hydrogen = 11.7; [1 mark]

- Ratio of carbon : oxygen : hydrogen is 5 : 1 : 10

**AND**

Therefore, the empirical formula is  $\text{C}_5\text{H}_{10}\text{O}$ ; [1 mark]

- Molecular formula =  $\text{C}_5\text{H}_{10}\text{O}$

**AND**

Because the empirical formula has a mass of 86 which corresponds to the  $M+1$  peak at  $m/z = 87$ ; [1 mark]

**[Total: 3 marks]**

- This question contains all the little tricks that examiners can do:

- Not giving you all of the percentages and expecting you to deduce the missing percentage
  - Percentage of hydrogen =  $100 - 69.7 - 18.6 = 11.7\%$
- Giving you the elements out of the standard, and almost expected, order
- Not giving you the specific mass of the unknown compound - although you should be able to determine this from the M+1 peak information
- To calculate the empirical formula:
- 

| Element             | C                          | O                          | H                           |
|---------------------|----------------------------|----------------------------|-----------------------------|
| Value from question | 69.7                       | 18.6                       | 11.7                        |
| $A_r$               | 12.0                       | 16.0                       | 1.0                         |
| Moles               | $\frac{69.7}{12.0} = 5.80$ | $\frac{18.6}{16.0} = 1.16$ | $\frac{11.7}{1.0} = 11.7$   |
| Ratio               | $\frac{5.80}{1.16} = 5$    | $\frac{1.16}{1.16} = 1$    | $\frac{11.7}{1.16} = 10.09$ |

- Therefore, the empirical formula is  $C_5H_{10}O$
- $C_5H_{10}O$  has a molar mass of  $(12.0 \times 5) + (1.0 \times 10) + 16.0 = 86.0$
- This means that  $C_5H_{10}O$  would have an M+1 peak at  $m/e = 87.0$

because the empirical formula has a mass of 86 which corresponds to the  $M+1$  peak at  $m/z = 87$ ; [1 mark]

**[Total: 3 marks]**

- This question contains all the little tricks that examiners can do:
  - Not giving you all of the percentages and expecting you to deduce the missing percentage
    - Percentage of hydrogen =  $100 - 69.7 - 18.6 = 11.7\%$
  - Giving you the elements out of the standard, and almost expected, order
  - Not giving you the specific mass of the unknown compound - although you should be able to determine this from the  $M+1$  peak information
- To calculate the empirical formula:

| Element             | C                          | O                          | H                           |
|---------------------|----------------------------|----------------------------|-----------------------------|
| Value from question | 69.7                       | 18.6                       | 11.7                        |
| $A_r$               | 12.0                       | 16.0                       | 1.0                         |
| Moles               | $\frac{69.7}{12.0} = 5.80$ | $\frac{18.6}{16.0} = 1.16$ | $\frac{11.7}{1.0} = 11.7$   |
| Ratio               | $\frac{5.80}{1.16} = 5$    | $\frac{1.16}{1.16} = 1$    | $\frac{11.7}{1.16} = 10.09$ |

- Therefore, the empirical formula is  $C_5H_{10}O$
- $C_5H_{10}O$  has a molar mass of  $(12.0 \times 5) + (1.0 \times 10) + 16.0 = 86.0$
- This means that  $C_5H_{10}O$  would have an  $M+1$  peak at  $m/e = 87.0$

2b

b) **Two** possible structures for compound **B** are:

- Pentan-2-one
- AND**
- Pentan-3-one; [1 mark]

Explanation:

- The (sharp) IR peak at  $1705\text{ cm}^{-1}$  corresponds to a carbonyl group /  $C=O$
- AND**
- Which suggests / could be an aldehyde or a ketone; [1 mark]
- The fragment ion peak at  $m/e = 28$  can only come from a  $C=O$  fragment

**AND**

Pentanal cannot (easily) form this fragment; [1 mark]

**[Total: 3 marks]**

- Using a combination of the molecular formula of  $C_5H_{10}O$  and the IR peak at  $1705\text{ cm}^{-1}$ , you should be able to determine pentanal, pentan-2-one and pentan-3-one as the only possible isomers
- The fragment ion peak at  $m / e = 28$  removes pentanal as a possibility because it cannot (easily) form that fragment ion

2c

c) The fragment responsible for the mass spectrum peak at  $m / e = 29$  is:

- $C_2H_5^+$ ; [1 mark]

How it can be used to distinguish between the two isomers:

- Pentan-2-one can only form one ( $C_2H_5^+$ ) fragment

**AND**

Pentan-3-one can only form two ( $C_2H_5^+$ ) fragments; [1 mark]

- Therefore, pentan-2-one will have a lower relative abundance of the ( $C_2H_5^+$ ) fragment

**OR**

Therefore, pentan-3-one will have a higher relative abundance of the ( $C_2H_5^+$ ) fragment; [1 mark]

**[Total: 3 marks]**

- **Remember:** A mass spectrum gives  $m / e$  data which can be used to identify potential fragment ions, but the size of the peaks can also give an indication of how much there is of each peak
  - Pentan-2-one can only form one peak at  $m / e = 29$
  - Due to its symmetry, pentan-3-one can essentially form 2 peaks at  $m / e = 29$
  - Therefore, you could expect the  $m / e = 29$  peak to be higher (potentially double) for pentan-3-one

3a

a) **One** possible identity of compound **C** is:

- Propanenitrile / propionitrile /  $\text{C}_2\text{H}_5\text{CN}$ ; [1 mark]

Explanation:

- The IR spectrum shows a peak at around  $2250\text{ cm}^{-1}$

**AND**

Which suggests a nitrile group /  $\text{C}\equiv\text{N}$ ; [1 mark]

- $\text{C}\equiv\text{N}$  has a mass of 26

**AND**

$55 - 26 = 29$ , which corresponds to an ethyl group; [1 mark]

**[Total: 3 marks]**

- This question asks for you to identify a chemical that is one of the less common structures in structure determination exam questions - a nitrile
  - To allow this question to be asked, the IR spectrum is actually edited to remove a sharp spike that appears around  $1750\text{ cm}^{-1}$  and would mislead you into thinking the unknown organic compound was a carbonyl
- One obvious peak on the IR spectrum is at around  $3000\text{ cm}^{-1}$ , which corresponds to C–H bonds
- The question advises to not use peaks below  $1500\text{ cm}^{-1}$ 
  - These peaks are in the fingerprint region and could only be used by comparison to a database, which would most likely be done using a computer
  - Also, the peak at around  $1450\text{ cm}^{-1}$  does not correspond to anything in the data table
- This leaves the peak at around  $2250\text{ cm}^{-1}$  which could easily be missed as it is not as large / obvious as you might normally expect
  - This peak corresponds to the triple bond of a nitrile group
- A nitrile group has a mass of 26
  - Removing this mass from the molecular ion peak =  $55 - 26 = 29$
  - You should recognise that 29 is a common fragment formed by an ethyl group/ion
- This leads to a final answer of propanenitrile /  $\text{C}_2\text{H}_5\text{CN}$

3b

b) **Two** ways that the  $\text{M}+1$  peak at  $m/e = 56$  could be formed are:

- Due to the carbon-13 isotope

**AND**

By the addition of a proton to the molecular ion; [1 mark]

**[Total: 1 mark]**

- You are expected to know that the most common reason for the M+1 peak is due to the natural abundance of the carbon-13 isotope
- However, you could be expected to deduce that the addition of a proton could also cause the M+1 peak
  - This happens quite frequently in nitriles according to scientific publications

3c

c) The expected elemental composition of an 8.0 g sample of compound **C** is:

- Percentage by mass of carbon =  $\frac{36}{55} \times 100 = 65.45\%$

**AND**

Percentage by mass of hydrogen =  $\frac{5}{55} \times 100 = 9.09\%$

**AND**

Percentage by mass of nitrogen =  $\frac{14}{55} \times 100 = 25.45\%$ ; [1 mark]

- Mass of carbon in 8 g sample =  $8 \times \frac{65.45}{100} = 5.24 \text{ g}$

**AND**

Mass of hydrogen in 8 g sample =  $8 \times \frac{9.09}{100} = 0.72 \text{ g}$

**AND**

Mass of nitrogen in 8 g sample =  $8 \times \frac{25.45}{100} = 2.04 \text{ g}$ ; [1 mark]

**[Total: 2 marks]**

- This question is basically testing your ability to do an empirical formula calculation backwards
  - A backwards empirical formula calculation is actually percentage by mass
- The percentages you calculate are then applied to the 8 g sample given in the question
  - You can check your answer is correct because the answer should add up to 8 g

3d

d) The reagents required for this conversion of compound **C** into compound **D** are:

- Hydrogen gas /  $\text{H}_2$  (g) **AND** nickel catalyst  
**OR**  
Lithium aluminium hydride /  $\text{LiAlH}_4$  **AND** dry ether; [1 mark]

Justification:

- The nitrile /  $\text{C}\equiv\text{N}$  peak is replaced by a peak at  $3300 - 3500\text{ cm}^{-1}$   
**AND**  
Which corresponds to N-H bonds; [1 mark]

[Total: 2 marks]

- The most likely reaction for any nitrile compound is reduction
  - You should know the reagents required for this reaction
    - Hydrogen gas /  $\text{H}_2$  (g) with a nickel catalyst is the most common answer to this question
    - Lithium aluminium hydride /  $\text{LiAlH}_4$  and dry ether is another possible answer, although this actually comes from the 7.6 Nitrogen Compounds A-level topic
- After stating the reagents, you have to use the IR spectrum to show that the reduction reaction has taken place
  - This would require the  $\text{C}\equiv\text{N}$  to break and therefore disappear from the IR spectrum
  - This would cause the formation of N-H bonds which would appear on the IR spectrum

4a

a) To calculate the molecular formula of alcohol **E**:

- Moles of carbon = 5.68  
**AND**  
Moles of hydrogen = 13.6  
**AND**  
Moles of oxygen = 1.14; [1 mark]

- Ratio of carbon : hydrogen : oxygen is 5 : 12 : 1

**AND**

Therefore, the empirical formula is  $\text{C}_5\text{H}_{12}\text{O}$ ; [1 mark]

- $M_r$  of the empirical formula =  $(5 \times 12.0) + (12.0 \times 1) + 16.0 = 88.0$

**AND**

Therefore, the molecular formula is  $\text{C}_5\text{H}_{12}\text{O}$ ; [1 mark]

**[Total: 3 marks]**

- Part (i) is a standard empirical formula calculation

•

| Element             | C                          | H                           | O                          |
|---------------------|----------------------------|-----------------------------|----------------------------|
| Value from question | 68.2                       | 13.6                        | 18.2                       |
| $A_r$               | 12.0                       | 1.0                         | 16.0                       |
| Moles               | $\frac{68.2}{12.0} = 5.68$ | $\frac{13.6}{1.0} = 13.6$   | $\frac{18.2}{16.0} = 1.14$ |
| Ratio               | $\frac{5.68}{1.14} = 4.98$ | $\frac{13.6}{1.14} = 11.93$ | $\frac{1.14}{1.14} = 1$    |

- Therefore, the empirical formula is  $\text{C}_5\text{H}_{12}\text{O}$

although this actually comes from the 7.6 Nitrogen Compounds A-level topic

- After stating the reagents, you have to use the IR spectrum to show that the reduction reaction has taken place
  - This would require the  $C \equiv N$  to break and therefore disappear from the IR spectrum
  - This would cause the formation of N-H bonds which would appear on the IR spectrum

4a

a) To calculate the molecular formula of alcohol E:

- Moles of carbon = 5.68

**AND**

Moles of hydrogen = 13.6

**AND**

Moles of oxygen = 1.14; [1 mark]

- Ratio of carbon : hydrogen : oxygen is 5 : 12 : 1

**AND**

Therefore, the empirical formula is  $C_5H_{12}O$ ; [1 mark]

- $M_r$  of the empirical formula =  $(5 \times 12.0) + (12.0 \times 1) + 16.0 = 88.0$

**AND**

Therefore, the molecular formula is  $C_5H_{12}O$ ; [1 mark]

**[Total: 3 marks]**

- Part (i) is a standard empirical formula calculation

•

| Element             | C                          | H                           | O                          |
|---------------------|----------------------------|-----------------------------|----------------------------|
| Value from question | 68.2                       | 13.6                        | 18.2                       |
| $A_r$               | 12.0                       | 1.0                         | 16.0                       |
| Moles               | $\frac{68.2}{12.0} = 5.68$ | $\frac{13.6}{1.0} = 13.6$   | $\frac{18.2}{16.0} = 1.14$ |
| Ratio               | $\frac{5.68}{1.14} = 4.98$ | $\frac{13.6}{1.14} = 11.93$ | $\frac{1.14}{1.14} = 1$    |

- Therefore, the empirical formula is  $C_5H_{12}O$
- **Careful:** Don't assume that the empirical formula is the molecular formula

- Use the molecular ion  $m/e$  value of 88 to determine if the empirical formula is the molecular formula
- In this instance, the molecular and empirical formulae are the same
- Without this check, you will lose a mark

4b

b) The information this tells us about the structure of alcohol **E** is:

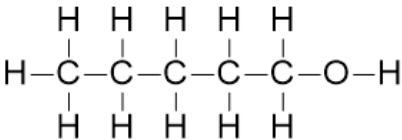
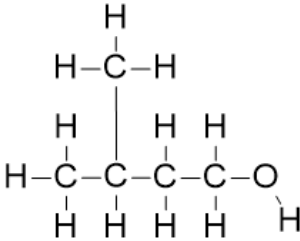
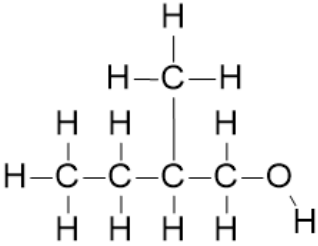
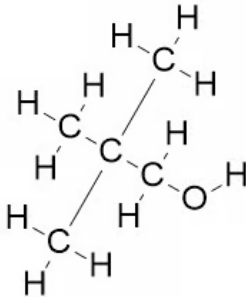
- It is a primary /  $1^\circ$  alcohol; [1 mark]

[Total: 1 mark]

- **Remember:**
  - Primary alcohols are oxidised to aldehydes and then carboxylic acids
  - Secondary alcohols are oxidised to ketones
  - Tertiary alcohols can't be further oxidised

4c

c) The fully displayed formulae and names of the **four** possible structural isomers that could be alcohol **E** are:

|                                                                                                                                           |                                                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p><b>AND</b></p> <p>Name = Pentan-1-ol; [1 mark]</p> |  <p><b>AND</b></p> <p>Name = 3-methylbutan-1-ol; [1 mark]</p> |
|  <p><b>AND</b></p>                                     |                                                              |

Name = 2-methylbutan-1-ol; [1 mark]

AND

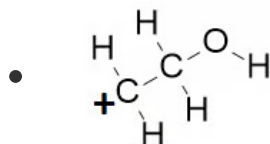
Name = 2,2-dimethylpropan-1-ol; [1 mark]

[Total: 4 marks]

- Take your time drawing the four structural isomers and naming them correctly:
  - From part (c), you are drawing structural isomers of primary alcohols so don't be tempted to draw secondary or tertiary structures
  - Always double check they are in fact isomers and not the same molecule
  - Take care with the naming rules as without a name you will not be awarded the full marks
- Apart from pentan-1-ol, you can also score the mark by giving a name that does not include the -1-, e.g. 3-methylbutanol
  - This is because the alcohol group is fixed at carbon 1
  - This means that the molecule is numbered from carbon-1, which is the carbon with the alcohol group

4d

d) The structure of the species that could give a major peak at  $m/e = 45$ .



AND

Positive charge included in the diagram; [1 mark]

[Total: 1 mark]

- Take your time deducing the fragment ion
  - It will be linked to one of the isomers that you have just drawn so you are best starting with the  $\text{CH}_2\text{OH}$  which is common to all of the isomers
  - This has a mass of 31
  - This leaves a mass of 14, which is a  $\text{CH}_2$  group
  - Therefore, the fragment is  $\text{CH}_2\text{CH}_2\text{OH}$
  - **Remember:** The question asks for the fragment ion which is  $\text{CH}_2\text{CH}_2\text{OH}^+$

4e

e) The identity of alcohol **E** is:

- 3-methyl-butan-1-ol

**AND**

This is the only alcohol with a branched chain that forms  $\text{CH}_2\text{OHCH}_2^+$  /  $\text{C}_2\text{H}_4\text{OH}^+$  / peak at 45.0; [1 mark]

[Total: 1 mark]

- There are three branched structural isomers:
  - 3-methylbutan-1-ol
  - 2-methylbutan-1-ol
  - 2,2-dimethylpropan-1-ol
- Only 3-methylbutan-1-ol can form a fragment ion with an  $m / e$  value of 45.0

5a

a)

i) To determine whether compound **F** is a carbohydrate according to the previous chemistry definition:

- Percentage of carbon = 55.81 (%)

**AND**

Moles of carbon = 4.65; [1 mark]

- Moles of hydrogen = 6.98

**AND**

Moles of oxygen = 2.34; [1 mark]

- Ratio of carbon : hydrogen: oxygen is 2 : 3 : 1

**AND**

Therefore, the empirical formula is  $\text{C}_2\text{H}_3\text{O}$ ; [1 mark]

- Compound **F** is not a carbohydrate according to the previous chemistry definition

**AND**

Because the number of hydrogen and oxygen atoms does not match the  $\text{H}_2\text{O}$  /  $(\text{H}_2\text{O})_n$  in the general formula; [1 mark]

ii) The answer to part (i) cannot be the molecular formula of compound **F** because:

- The number of hydrogens is not possible for a complete / fully bonded molecule
- OR**

The molecules possible with two carbon atoms and one oxygen atom have an incorrect number of hydrogen atoms; [1 mark]

Plus any **three** of the following:

•

| Displayed formula                                                                                                                                           | Justification, in terms of hydrogen atoms |            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|------------|
| $\begin{array}{c} \text{H} \\   \\ \text{C}=\text{C}=\text{O} \\   \\ \text{H} \end{array}$                                                                 | Not enough hydrogen atoms                 | ; [1 mark] |
| $\begin{array}{c} \text{H} \\   \\ \text{H}-\text{C}-\text{C}=\text{O} \\   \quad   \\ \text{H} \quad \text{H} \end{array}$                                 | Too many hydrogen atoms                   | ; [1 mark] |
| $\begin{array}{c} \text{O} \\ / \quad \backslash \\ \text{H}-\text{C}-\text{C}-\text{H} \\   \quad   \\ \text{H} \quad \text{H} \end{array}$                | Too many hydrogen atoms                   | ; [1 mark] |
| $\begin{array}{c} \text{H} \quad \text{H} \\   \quad   \\ \text{H}-\text{C}-\text{O}-\text{C}-\text{H} \\   \quad   \\ \text{H} \quad \text{H} \end{array}$ | Too many hydrogen atoms                   | ; [1 mark] |
| $\begin{array}{c} \text{H} \quad \text{H} \\   \quad   \\ \text{H}-\text{C}-\text{C}-\text{O}-\text{H} \\   \quad   \\ \text{H} \quad \text{H} \end{array}$ | Too many hydrogen atoms                   | ; [1 mark] |

**[Total: 8 marks]**

- For part (i)
  - This question gives you the elements out of the standard, and almost expected, order
  - You also have to calculate the missing percentage:
    - Percentage of hydrogen =  $100 - 6.98 - 37.21 = 55.81 \%$

- To calculate the empirical formula:

- 

| Element             | C                           | H                          | O                           |
|---------------------|-----------------------------|----------------------------|-----------------------------|
| Value from question | 55.81                       | 6.98                       | 37.21                       |
| $A_r$               | 12.0                        | 1.0                        | 16.0                        |
| Moles               | $\frac{55.81}{12.0} = 4.65$ | $\frac{6.98}{1.0} = 6.98$  | $\frac{37.21}{16.0} = 2.34$ |
| Ratio               | $\frac{4.65}{2.34} = 1.99$  | $\frac{6.98}{2.34} = 2.98$ | $\frac{2.34}{2.34} = 1$     |

- Therefore, the empirical formula is  $C_2H_3O$
- It is the  $H_2O$  portion of the general formula  $C_m(H_2O)_n$  that is important
  - The number of carbon atoms,  $m$ , is independent of the hydrogen and oxygen atoms
  - The  $(H_2O)_n$  portion shows that there must always be a ratio of 2 hydrogen atoms : 1 oxygen atom
  - The  $H_3O$  portion of the empirical formula cannot satisfy this ratio
  - Therefore, compound **F** cannot be a carbohydrate according to the previous chemistry definition